

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 22

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Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.3, 16.4 (fundamental theorem of line integrals and Green's theorem)
- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104

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ALVARO: Start the recording!



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Green's Theorem



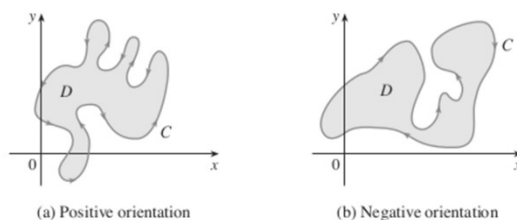
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About Parametrized Curves

A parametrized curve is called closed or a loop if it starts and ends at the same point.

A closed curve is simple if its intersection with itself is at the start and the end, and nowhere else.

A simple closed curve on the plane is positively oriented if it goes counterclockwise. Otherwise it is negatively oriented.



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Green's Theorem

Green's Theorem

Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C . If P and Q have continuous partial derivatives on an open region that contains D , then

$$\int_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

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Example: Evaluate $\int_C x^4 dx + xy dy$ where C is the triangular curve along the segments from $(0,0)$ to $(1,0)$, from $(1,0)$ to $(0,1)$, and from $(0,1)$ to $(0,0)$.

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[Extra Space]

Example: Evaluate $\int_C x^4 dx + xy dy$ where C is the triangular curve along the segments from $(0,0)$ to $(1,0)$, from $(1,0)$ to $(0,1)$, and from $(0,1)$ to $(0,0)$.

8

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$$\int_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Notes: Sometimes we write $\oint_C \dots$ to denote we are integrating along a closed curve.

If $F = (P, Q)$ is conservative, then $F = \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$ and therefore

$$\frac{\partial P}{\partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial Q}{\partial x}$$

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Example: Evaluate $\oint_C (3y - e^{\sin(x)})dx + (7x + \sqrt{y^4 + 1})dy$
where C is the circle of radius 3 centered at the origin.

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[Extra Space]

Example: Evaluate $\oint_C (3y - e^{\sin(x)})dx + (7x + \sqrt{y^4 + 1})dy$
 where C is the circle of radius 3 centered at the origin.

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Green's Theorem

Green's Theorem

Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C . If P and Q have continuous partial derivatives on an open region that contains D , then

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$$

$$\int_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$$P(x, y) = 0 \quad P(x, y) = -y \quad P(x, y) = -\frac{1}{2}y$$

$$Q(x, y) = x \quad Q(x, y) = 0 \quad Q(x, y) = \frac{1}{2}x$$

$$A = \oint_C x dy = -\oint_C y dx = \frac{1}{2} \oint_C x dy - y dx$$

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Example: Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.



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Example: Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

[Extra Space]



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Example: Find the area of the triangle with vertices $(0,0)$, $(2,0)$, $(2,3)$.

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Example: Find the area of the triangle with vertices $(0,0)$, $(2,0)$, $(2,3)$.

[Extra Space]

16

Questions?

17



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Curl and Divergence

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Curl

If $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and the partial derivatives of P , Q , and R all exist, then the curl of $\mathbf{F}$ is the vector field on $\mathbb{R}^3$ defined by

1

$$\text{curl } \mathbf{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k}$$

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

$$= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k}$$

$$= \text{curl } \mathbf{F}$$

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$$

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Example: Find the curl of $F(x, y, z) = (xz, xyz, -y^2)$.



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Example: Find the curl of $F(x, y, z) = (xz, xyz, -y^2)$.

[Extra Space]



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Curl

If $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and the partial derivatives of P , Q , and R all exist, then the curl of $\mathbf{F}$ is the vector field on $\mathbb{R}^3$ defined by

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$$\text{curl } \mathbf{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k}$$

3 Theorem

If f is a function of three variables that has continuous second-order partial derivatives, then

$$\text{curl}(\nabla f) = \mathbf{0}$$

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Curl

3 Theorem

If f is a function of three variables that has continuous second-order partial derivatives, then

$$\text{curl}(\nabla f) = \mathbf{0}$$

Proof

We have

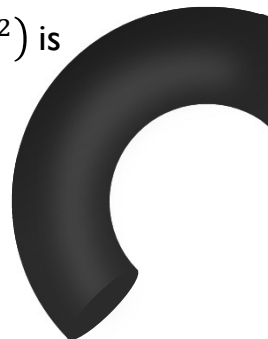
$$\begin{aligned} \text{curl}(\nabla f) &= \nabla \times (\nabla f) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix} \\ &= \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y} \right) \mathbf{i} + \left(\frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z} \right) \mathbf{j} + \left(\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \right) \mathbf{k} \\ &= 0 \mathbf{i} + 0 \mathbf{j} + 0 \mathbf{k} = \mathbf{0} \end{aligned}$$

by Clairaut's Theorem.

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Example: Show that the vector field $F(x, y, z) = (xz, xyz, -y^2)$ is NOT conservative.

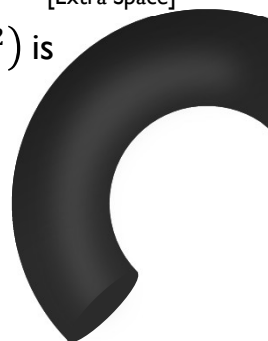
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Example: Show that the vector field $F(x, y, z) = (xz, xyz, -y^2)$ is NOT conservative.

[Extra Space]

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Curl

3 Theorem

If f is a function of three variables that has continuous second-order partial derivatives, then

$$\text{curl}(\nabla f) = \mathbf{0}$$

4 Theorem

If $\mathbf{F}$ is a vector field defined on all of $\mathbb{R}^3$ whose component functions have continuous partial derivatives and $\text{curl } \mathbf{F} = \mathbf{0}$, then $\mathbf{F}$ is a conservative vector field.

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Example: Show that the vector field

$F(x, y, z) = (y^2 z^3, 2xyz^3, 3xy^2 z^2)$ is conservative.

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[Extra Space]

Example: Find a function $f(x,y,z)$ such that its gradient is given by

$$F(x, y, z) = (y^2 z^3, 2xyz^3, 3xy^2 z^2).$$

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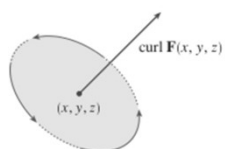
[Extra Space]

Example: Find a function $f(x,y,z)$ such that its gradient is given by

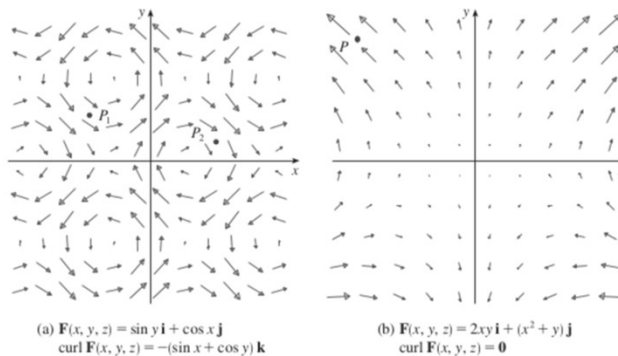
$$F(x, y, z) = (y^2 z^3, 2xyz^3, 3xy^2 z^2).$$

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Curl



Velocity fields in fluid flow. (Only the part of $\mathbf{F}$ in the xy -plane is shown; the vector field looks the same in all horizontal planes because $\mathbf{F}$ is independent of z and the z -component is 0.)



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Divergence

If $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and $\partial P/\partial x$, $\partial Q/\partial y$ and $\partial R/\partial z$ exist, then the **divergence** of $\mathbf{F}$ is the function of three variables defined by

9

$$\text{div } \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

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Divergence

If $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and $\partial P/\partial x$, $\partial Q/\partial y$ and $\partial R/\partial z$ exist, then the **divergence of $\mathbf{F}$** is the function of three variables defined by

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$$\operatorname{div} \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

$$\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F}$$

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Example: Find the divergence of $F(x, y, z) = (xz, xyz, -y^2)$.

34

Example: Find the divergence of $F(x, y, z) = (xz, xyz, -y^2)$.

[Extra Space]

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Divergence

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$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F}$$

11 Theorem

If $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and P , Q , and R have continuous second-order partial derivatives, then

$$\operatorname{div} \operatorname{curl} \mathbf{F} = 0$$

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Divergence

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F}$$

Theorem

If $\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$ is a vector field on $\mathbb{R}^3$ and P , Q , and R have continuous second-order partial derivatives, then

$$\operatorname{div} \operatorname{curl} \mathbf{F} = 0$$

Proof

Using the definitions of divergence and curl, we have

$$\begin{aligned} \operatorname{div} \operatorname{curl} \mathbf{F} &= \nabla \cdot (\nabla \times \mathbf{F}) \\ &= \frac{\partial}{\partial x} \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \\ &= \frac{\partial^2 R}{\partial x \partial y} - \frac{\partial^2 Q}{\partial x \partial z} + \frac{\partial^2 P}{\partial y \partial z} - \frac{\partial^2 R}{\partial y \partial x} + \frac{\partial^2 Q}{\partial z \partial x} - \frac{\partial^2 P}{\partial z \partial y} \\ &= 0 \end{aligned}$$

because the terms cancel in pairs by Clairaut's Theorem.

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Example: Show that the vector field $F(x, y, z) = (xz, xyz, -y^2)$ is not a curl, i.e., there is no vector field G such that $F = \operatorname{curl} G$.

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[Extra Space]

Example: Find the divergence of $F(x, y, z) = (xz, xyz, -y^2)$.

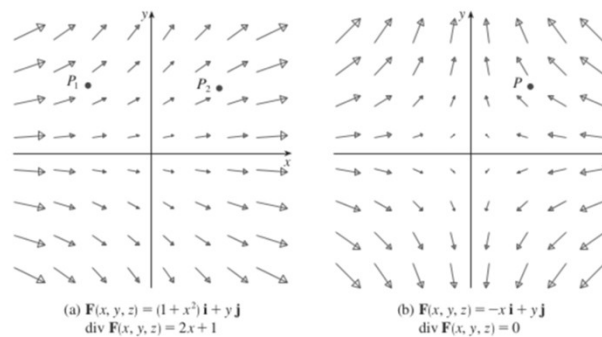
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Divergence

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F}$$

Again, the reason for the name *divergence* can be understood in the context of fluid flow. If $\mathbf{F}(x, y, z)$ is the velocity of a fluid (or gas), then $\operatorname{div} \mathbf{F}(x, y, z)$ represents the net rate of change (with respect to time) of the mass of fluid (or gas) flowing from the point (x, y, z) per unit volume. In other words, $\operatorname{div} \mathbf{F}(x, y, z)$ measures the tendency of the fluid to diverge from the point (x, y, z) . If $\operatorname{div} \mathbf{F} = 0$, then $\mathbf{F}$ is said to be **incompressible**.

Velocity fields in fluid flow. (Only the part of $\mathbf{F}$ in the xy -plane is shown; the vector field looks the same in all horizontal planes because $\mathbf{F}$ is independent of z and the z -component is 0.)



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Laplace

Another differential operator occurs when we compute the divergence of a gradient vector field ∇f .

If f is a function of three variables, we have

$$\operatorname{div}(\nabla f) = \nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

and this expression occurs so often that we abbreviate it as $\nabla^2 f$. The operator

$$\nabla^2 = \nabla \cdot \nabla$$

is called the **Laplace operator** because of its relation to **Laplace's equation**

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

$$\mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$$

$$\nabla^2 \mathbf{F} = \nabla^2 P \mathbf{i} + \nabla^2 Q \mathbf{j} + \nabla^2 R \mathbf{k}$$

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Vector Form of Green's Theorem

Consider $\mathbf{F} = (P, Q)$ satisfying Green's Theorem hypotheses.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C P dx + Q dy$$

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Vector Form of Green's Theorem

Consider $\mathbf{F} = (P, Q)$ satisfying Green's Theorem hypotheses.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C P dx + Q dy$$

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P(x, y) & Q(x, y) & 0 \end{vmatrix} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k}$$

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Vector Form of Green's Theorem

Consider $\mathbf{F} = (P, Q)$ satisfying Green's Theorem hypotheses.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C P dx + Q dy \quad \text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P(x, y) & Q(x, y) & 0 \end{vmatrix} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k}$$

Therefore:

$$(\text{curl } \mathbf{F}) \cdot \mathbf{k} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k} \cdot \mathbf{k} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

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Vector Form of Green's Theorem

Consider $\mathbf{F} = (P, Q)$ satisfying Green's Theorem hypotheses.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C P dx + Q dy \quad (\text{curl } \mathbf{F}) \cdot \mathbf{k} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k} \cdot \mathbf{k} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

Then Green's Theorem says:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C \mathbf{F} \cdot \mathbf{T} ds = \iint_D (\text{curl } \mathbf{F}) \cdot \mathbf{k} dA$$

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Vector Form of Green's Theorem

Consider $\mathbf{F} = (P, Q)$ satisfying Green's Theorem hypotheses.

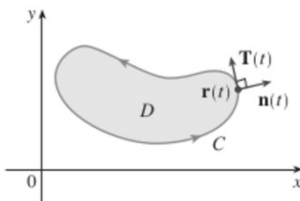
$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C P dx + Q dy$$

$$\mathbf{r}(t) = x(t) \mathbf{i} + y(t) \mathbf{j} \quad a \leq t \leq b$$

$$\mathbf{T}(t) = \frac{x'(t)}{|\mathbf{r}'(t)|} \mathbf{i} + \frac{y'(t)}{|\mathbf{r}'(t)|} \mathbf{j}$$

$$\mathbf{n}(t) = \frac{y'(t)}{|\mathbf{r}'(t)|} \mathbf{i} - \frac{x'(t)}{|\mathbf{r}'(t)|} \mathbf{j}$$

Then Green's Theorem alternatively says:



$$\oint_C \mathbf{F} \cdot \mathbf{n} ds = \iint_D \text{div } \mathbf{F}(x, y) dA$$

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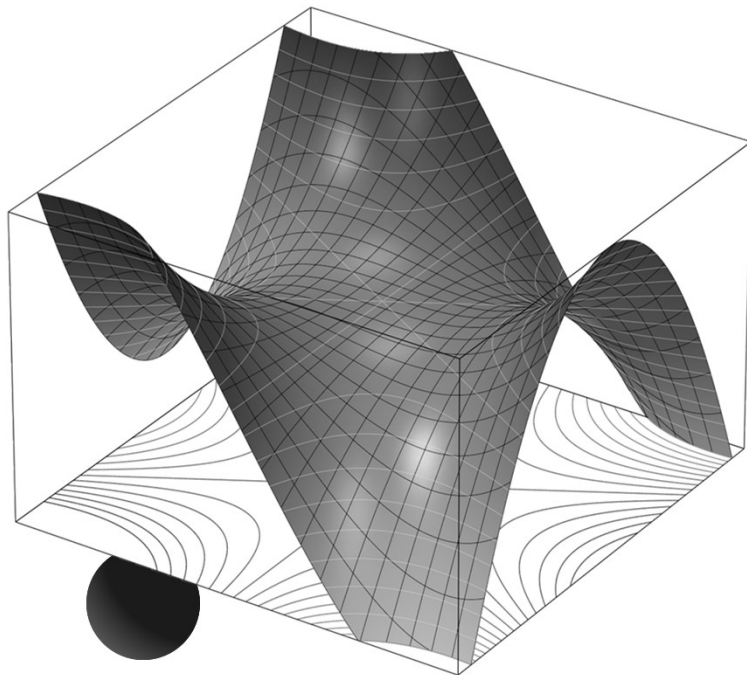
Questions?



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Thank you

Until next time.



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